Solving Problems Involving Algebraic Expressions and Formulas

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Solving Using a Formula

To **solve a problem using a formula**, identify the given quantities, select the appropriate formula, substitute the known values, and calculate the unknown quantity.

Memory Aid: G–F–S–A

G **G**iven quantities

F **F**ormula to use

S **S**ubstitute the values

A **A**nswer with the unit

Finding an Unknown Variable

To **find an unknown variable in a formula**, use the properties of equality to isolate the variable.

Perform the **same operation on both sides** of the equation.

Isolating the Variable

$$8n = 96$$

Divide both sides by 8.

$$\frac{8n}{8} = \frac{96}{8}$$

$$n = 12$$

Unit of Measurement

Always include the correct **unit of measurement** when the problem involves quantities such as length, area, distance, time, or money.

Examples of Units

Quantity	**Unit**
Length	cm, m
Area	cm^2, m^2
Distance	km
Time	hours, minutes
Money	₱

Examples

Example 1

A pen costs ₱50. The total cost is $C = 50x$, where x is the number of pens. Find the total cost of 4 pens.

$$C = 50(4)$$

$$C = 200$$

The total cost is ₱200.

Example 2

The area of a rectangle is $A = lw$. Find the area of a rectangle with length 12 m and width 5 m.

$$A = (12)(5)$$

$$A = 60$$

The area is $60\ \mathrm{m}^2$.

Example 3

The distance traveled is $d = rt$. A car travels 240 km at 60 km/h. Find the travel time.

$$240 = 60t$$

$$\frac{240}{60} = \frac{60t}{60}$$

$$t = 4$$

The travel time is 4 hours.

Example 4

A worker earns $E = 100h$ pesos for h hours of work. If the worker earns ₱700, how many hours did the worker work?

$$700 = 100h$$

$$\frac{700}{100} = \frac{100h}{100}$$

$$h = 7$$

The worker worked 7 hours.

Practice Exercises

Practice 1

Solve the problem.

The expression $3x + 8$ represents the total number of stickers Carlo has after receiving 8 additional stickers. Evaluate the expression when $x = 12$.

Practice 2

Solve the problem.

The area of a rectangle is given by $A = lw$. A rectangular garden is 15 m long and 8 m wide. Find its area.

Practice 3

Solve the problem.

The total fare for a group of students is represented by $F = 75n$, where n is the number of students. If the total fare is ₱900, find the number of students.

Practice 4

Solve the problem.

A worker earns ₱120 per hour. His total earnings are given by $E = 120h$. If he earns ₱960, how many hours did he work?

Practice 5

Solve the problem.

A water container holds $V = 12t$ liters of water after being filled at a constant rate of 12 liters per minute for t minutes. How much water will it contain after 7 minutes? If the container has a capacity of 120 liters, how many minutes will it take to fill the container at this rate?

Activities

Activity 1

Solve the problem.

A notebook costs ₱380. If Ana buys x notebooks, the total cost is represented by $C = 380x$. Find the total cost if she buys 7 notebooks.

Activity 2

Solve the problem.

The perimeter of a rectangle is given by $P = 2l + 2w$. Find the perimeter of a rectangle whose length is 16 cm and width is 7 cm.

Activity 3

Solve the problem.

A bus travels at a constant speed. The distance traveled is given by $d = rt$, where r is the speed and t is the travel time. Find the distance traveled at 45 km/h for 3 hours.

Activity 4

Solve the problem.

The formula for the area of a triangle is $A = \frac{1}{2}bh$. Find the area of a triangle with a base of 20 cm and a height of 12 cm.

Activity 5

Solve the problem.

The formula for the perimeter of a square is $P = 4s$, where s is the side length. Find the side length of a square whose perimeter is 68 cm.

Thank you!